

Admire Theophilus Madyira

Greater Seattle Area

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EXPERIENCE

Microsoft Corporation / Redmond, WA

Aug 2025 - Present

Software Engineer

- Halved Edge Update latency and doubled user bandwidth savings by cutting payload sizes from **400MB to 200MB** via computing differential patches using hash-based fingerprints; shipped to **1B+ devices** after end-to-end ownership across client logic in **C++**, packaging & code signing in **bash & Python**, release & flighting API changes in **C#** and rollout monitoring via **Kusto QL** dashboards.
- Built AI agents wired into Copilot & Edge build & release pipelines (Canary, Dev, Beta & Stable) that monitor Kusto/ADX dashboards and alerts for build health, promotion latency, failure rates, and fleet update telemetry, auto-triaging build/test failures during release rotations and cutting incident triage from **30+ min to ~5 min**.
- Owned Linux incident response: resolved **10 incidents** and root-caused failures impacting **~20,000 devices** within three months. Became the first person to ship a configuration validation using **YAML & Docker** into an almost decade old Smoke test pipeline to avoid issue resurgence.
- Designed & implemented code check-in policies that now validate **~400 PRs/week** expanding check-in guardrails to block stale compliance assessments and detecting API changes lacking privacy review; integrated PR context into privacy reviewer tasks, eliminating navigation overhead for product managers. Earned **full ownership** of end-to-end automation of the Edge and Copilot compliance pipeline **within 3 months** of joining.

Software Engineer Intern

May 2024 - Aug 2024

- Designed and shipped a **novel LLM council framework** for model vulnerability detection, achieving **~90% accuracy** in identifying prompt-injection, data-leakage, and jailbreak exploits in Copilot; delivered **one month ahead of deadline**.
- Requested a second internship project and built an **AI agent** for bug-fixing and browser crash analysis before the prevalence of AI coding tools, reducing engineers' triage time by **~50%** and securing a return offer before the internship ended.

Explore Intern

May 2023 - Aug 2023

- Optimized Microsoft Edge **search results algorithm** and **improved latency** in **C++**, contributing to a **~5% lift** in user retention.
- Conducted an **A/B experiment** validating net positive impact on Microsoft Bing's revenue.

Columbia University | New York, NY

June 2022 - Aug 2022

Machine Learning Engineer Intern

- Researched program synthesis for **autonomous systems** with Prof. Mark Santolucito, focused on automated controller generation and evaluation in the CARLA simulator.
- Trained RL motion-control policies (**PPO, SAC, DQN**) for autonomous vehicles in the CARLA simulator, improving trajectory stability and decision accuracy across diverse driving scenarios.

Google | Remote

June 2021 - May 2022

Full Stack Engineer Apprentice

- Selected and participated in a **10 month** long Google Africa Developer Scholarship program under expert Software Engineers from Google ending with a capstone project.

TECHNICAL SKILLS

Languages: C++, Python, Java, TypeScript, JavaScript, Bash, PowerShell, C#, SQL.

Frameworks: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Kubernetes, Docker, Terraform, GitHub Actions, Kusto/ADX.

Skills: Reinforcement Learning (PPO, SAC, DQN), NLP, RAG, LLM Fine-Tuning, MLOps, LLM Ops, AI Agents, CI/CD, Infrastructure as Code (IaC), SRE, Observability, Incident Management, System Design, Distributed Systems, Ultra Low Latency Optimization.

SELECT PROJECTS

- **QuantML** — Built an ML pipeline for systematic trading — **scikit-learn + PyTorch** models (**0.58 AUC**, walk-forward validated) and a **LoRA-fine-tuned** transformer for sentiment scoring — containerized with **Docker**, deployed to Azure, and gated by an **MLOps** evaluation harness with live paper-trading execution."
- **TenantIQ** — AI leasing agent (TypeScript/Node) with a scikit-learn risk-scoring model (**0.825 ROC-AUC**), served via **FastAPI/Docker** and gated by an **MLflow** evaluation harness, with CI automating retrain/eval.

EDUCATION

Columbia University - MSc. in Computer Science, Machine Learning Specialization

Expected May 2028

Amherst College - BA in Computer Science & Mathematics

May 2025

AWARDS & HONORS

- Rhodes Scholarship Finalist (2025) – Selected as **1 of 10** national finalists competing for the prestigious post-graduate scholarship at Oxford University.
- 2× Walker Award in Mathematics & Statistics, Amherst College (2023, 2025) - For achievement in Mathematics & Statistics
- 4× Koenig Scholarship, Amherst College (2021–2025) - Full merit scholarship for **only five admits**, renewed annually.
- Presidential & National Scholarship, Gov. of Zimbabwe (2020) - Full study-abroad funding for nationwide **top 10 high achieving** underprivileged students after ranking **1st of 50,000+** students on national A-Level exams.